Waseem Raza, PhD

Data Scientist | Machine Learning, Predictive Analytics & Cloud Data

Python | SQL | Azure Databricks | Statistical Modeling | Operational Analytics

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LinkedIn | GitHub | Google Scholar | Coursera Courses | Portfolio

Professional Profile

Data Scientist and Machine Learning Engineer with a PhD in Electrical and Computer Engineering and 7+ years of combined applied research, engineering, and data-driven problem-solving experience. Deliver statistical and machine-learning analysis of complex operational data, from extracting and validating records to feature engineering, model evaluation, and stakeholder communication. At AT&T, improved automated E911 root-cause classification performance from approximately 20% to 80%; at Impac, built Python/Azure data workflows that reduced data-preparation time by 40%. Hands-on with Python, SQL, scikit-learn, PyTorch, Azure Databricks, Synapse, Azure Data Factory, model validation, anomaly detection and cross-domain analytics. Interested in applying these methods to freight operations, asset reliability, and process improvement.

Technical Competencies

Data Science & Statistics: Python, SQL, Pandas, NumPy, SciPy, feature engineering, exploratory analysis, statistical testing, regression, classification, clustering, anomaly detection, time-series analysis, cross-validation, error analysis and experimental design

ML & AI Frameworks: scikit-learn, PyTorch, TensorFlow/Keras, LightGBM, XGBoost, neural networks, transfer learning, model selection, imbalanced learning, interpretability, model robustness and domain adaptation

Cloud Data Engineering: Azure Databricks, Apache Spark/PySpark, Azure Data Factory, Synapse Analytics, Azure Functions, Python/SQL ETL and ELT, ingestion, transformations, data quality, validation, pipeline troubleshooting and Azure DevOps CI/CD

Visualization & Reporting: Matplotlib, Plotly, Pandas, Jupyter, analytical reports, operational diagnostics, metrics and communicating model findings to technical and nontechnical stakeholders

Software & MLOps: Git, GitHub Actions, Linux, Docker, REST APIs, reproducible experiments, model evaluation, deployment-oriented workflows, basic Grafana exposure and systematic troubleshooting

Applied Analytics & Optimization: Network event data, reliability analysis, anomaly detection, large-scale logs, simulation, optimization, data-efficient learning and decision support for operational systems

Generative AI: LLM-enabled analytics, retrieval-augmented generation (RAG), embeddings, LangChain, LangGraph and structured extraction; applied research and portfolio workflows

Delivery and Collaboration

- **Business problem framing:** Translate ambiguous operational questions into measurable analytical objectives, defensible baselines and actionable results.
- **Cross-functional collaboration:** Work with network and software engineers, geologists, researchers and service-assurance stakeholders to define data requirements and explain findings.
- **Reliable experimentation:** Prioritize reproducible data processing, data-quality checks, cross-validation, failure analysis and clear interpretation of model limitations.
- **Communication:** Summarize complex modeling choices and results for mixed technical audiences through reports, plots and documented comparisons.

Professional Experience

Machine Learning Engineer **02/2025 - Present**

AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Develop and evaluate ML models for critical E911 service-assurance event data, improving automated root-cause classification performance from approximately 20% to 80% and enabling faster analysis of operational issues.
- Engineered text-feature and domain-adaptive classification workflows, handling label imbalance, data-quality differences and distribution shifts across operational domains.
- Analyzed work-order descriptions, network context and location-linked records to identify failure patterns; explored GenAI-assisted diagnostics in service-assurance prototypes.
- Built repeatable model-validation and error-analysis procedures to compare precision, recall, failure modes and cross-domain generalization for high-impact workflows.
- Collaborated with network engineers, software teams and public-safety stakeholders to turn operational questions into measurable ML tasks and communicate data-backed findings.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**

Impac Exploration Services *Weatherford, OK*

- Built an LLM-enabled analytical pipeline for extracting and interpreting information from geochemical well data.
- Developed Python/SQL-supported ingestion, cleansing, transformation and validation workflows with Azure Databricks, Azure Functions and Synapse Analytics.
- Automated ETL processing for large exploration datasets, reducing data-preparation time by 40% and improving reproducibility.
- Implemented cross-validation, data-quality, and consistency checks that improved reliability of analytical outputs by 20%.
- Collaborated with geologists to convert domain requirements and technical information into AI-ready datasets and actionable analytical outputs.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed more than 500K LTE/5G diagnostic logs with statistical and ML methods, identifying anomalous patterns and reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Advanced-Degrees Intern **06/2022 - 08/2022**
AT&T Labs *Bedminster, NJ*

- Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%.
- Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.
- Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.

Graduate Research Assistant **09/2019 - 07/2025**
AI4Networks, University of Oklahoma *Tulsa, OK*

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*
University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*
University of Engineering and Technology, Taxila, Pakistan

Selected Projects & Research

Projects & Applied Research

AI-Ready Lakehouse Pipeline *Portfolio Project | 2024*

Designed an AI-ready lakehouse pipeline that prepares structured and semi-structured data for analytics and downstream machine-learning workloads. Implemented repeatable ingestion, cleansing, transformation, validation, and scalable processing patterns suitable for reproducible cloud data workflows.

Tools & Methods: Azure Databricks, Apache Spark, PySpark, ETL/ELT, Data Ingestion, Schema Transformation, Data Validation, Feature Preparation, Cloud Data Engineering

Financial ML and Forecasting *Portfolio Project | 2024-2025*

Developed machine-learning workflows for structured credit-risk and loan-default datasets, together with time-series forecasting experiments. Applied data preparation, feature engineering, model comparison, cross-validation, and predictive-performance analysis to structured financial datasets.

Tools & Methods: Python, scikit-learn, Classification, Credit Risk, Loan Default Modeling, Time-Series Forecasting, Feature Engineering, Cross-Validation, Model Evaluation

Wearable Intelligence for Health Monitoring *Research Project | 2024-2026*

Developed deep-learning workflows for recognizing medication-taking and dietary activities from smartwatch accelerometer and gyroscope signals in free-living settings. Evaluated windowing, feature extraction, CNN/LSTM-style sequence models, cross-user generalization, and recall-oriented performance for health-monitoring applications.

Tools & Methods: Wearable AI, Smartwatch IMU, Time Series, Deep Learning, CNN/LSTM, Feature Engineering, Cross-User Evaluation, Activity Recognition, Healthcare AI

Agentic AI and RAG Applications *Portfolio Project | 2026*

Built grounded GenAI workflows that combine document retrieval, vector search, stateful orchestration, tool-using agents, memory, evaluation, and multi-agent handoffs. Emphasized controllable execution, traceable context flow, and reliable responses for multi-step tasks rather than single unconstrained model calls.

Tools & Methods: Python, LangGraph, LangChain, CrewAI, AutoGen, RAG, Vector Databases, Embeddings, Tool Calling, Agent Memory, Evaluation

Propagation Models & Digital Twins *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

Tools & Methods: Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020-2024*

Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.

Tools & Methods: Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

System Design Case-Study Portfolio *Guided Project | 2026*

Developed architecture case studies that translate ambiguous product requirements and traffic assumptions into service boundaries, APIs, data models, storage choices, caching strategies, and reliability targets. Documented scalability, consistency, capacity, and failure-mode tradeoffs to support production-oriented system design decisions.

Tools & Methods: System Design, REST APIs, Distributed Systems, Data Modeling, Caching, Scalability, Reliability, Capacity Planning, Architecture Tradeoffs

Publications

Real-World Medication Adherence Monitoring Using Hybrid Neural Networks and Smartwatches *2026*
HealthINF | Accepted

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks *2025*
Sensors 2025

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin *2024*
IEEE PIMRC

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks *2024*
IEEE SmartNets

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm) *2022*
IEEE Network

Towards Positioning Error Impact Characterization and Minimization in User-Centric RAN *2022*
IEEE WCNC

Unification of RF Energy Harvesting Schemes under Mixed Rayleigh-Rician Fading Channels *2020*
AEÜ - International Journal of Electronics and Communications

Certifications & Professional Development

1. Generative AI Applications with Amazon Bedrock — Coursera | Completed | August 27, 2026 | Certificate

Focus: Covers development of generative-AI applications on Amazon Bedrock, including foundation-model access, prompt-based application logic, model interaction, and managed AWS services. Emphasizes practical cloud patterns for building and operating LLM-powered applications.

Tools & Skills: Amazon Bedrock, foundation models, prompt engineering, AWS generative-AI services, LLM application development, cloud AI workflows

2. Agentic AI with LangChain and LangGraph — Coursera | Completed | February 9, 2026

Focus: Covers design of stateful agentic applications using LangChain and LangGraph, including graph-based orchestration, tool use, memory and state, routing, and multi-step workflows. Emphasizes controllable agent execution and composition of reliable LLM-driven processes.

Tools & Skills: LangChain, LangGraph, tool-calling, state and memory, workflow orchestration, routing, agent design

3. Machine Learning — Stanford Online / Coursera | Completed | March 29, 2020

Focus: Covers foundational machine-learning methods for supervised and unsupervised problems, including regression, classification, model evaluation, regularization, clustering, and practical model-selection considerations. Emphasizes translating data into predictive models while controlling generalization error.

Tools & Skills: supervised learning, regression, classification, regularization, clustering, model evaluation, bias-variance tradeoffs, machine-learning workflows

4. Introduction to Data Analytics in Google Cloud — Coursera | Completed | July 24, 2024

Focus: Introduces cloud-native data analytics in Google Cloud, covering the analytics lifecycle, data sources, storage and processing concepts, and how cloud services support scalable analysis. Emphasizes organizing data work to produce reliable analytical insight.

Tools & Skills: Google Cloud, cloud analytics, data lifecycle, data-ingestion concepts, storage and processing, analytical workflows, data-driven decision support

5. Introduction to Deep Learning & Neural Networks with Keras — IBM / Coursera | Completed | August 12, 2026

Focus: Introduces deep-learning and neural-network fundamentals using Keras, covering network architecture, training, validation, and model improvement. Connects concepts such as activation functions and optimization with practical neural-network implementation and evaluation.

Tools & Skills: Keras, deep learning, neural networks, model training, validation, activation functions, optimization, model evaluation